RIT*: Riemannian Informed Trees for Cost-Adaptive Optimal Motion Planning

Muhayy Ud Din, Ahmed Nadar, Jan Rosell, and Irfan Hussain

***Abstract*—Finding optimal collision-free paths in high-dimensional configuration spaces remains challenging. Asymptotically optimal planners such as batch-informed planners accelerate convergence by restricting sampling to an informed subset, but define this subset using Euclidean distance. This leads to inefficient sampling when motion costs are anisotropic or spatially varying (e.g., due to joint inertia or obstacle proximity), as the Euclidean informed set over-approximates the true cost-relevant region. Moreover, existing methods cannot learn a suitable cost metric during planning. To address these issues, we present RIT*: Riemannian Informed Trees, a method that substitutes Euclidean primitives in the batch-informed framework with their Riemannian counterparts, resulting in a more compact, cost-consistent sampling region. We additionally introduce a Collision-Adaptive Riemannian Metric (CARM), which updates the cost landscape online based on collision feedback, eliminating the need for prior knowledge of the environment. Experiments conducted in environments ranging from 2-D setups to 6-DOF manipulation tasks demonstrate that RIT* achieves higher-quality solutions than existing state-of-the-art planners. Experiments across four benchmarks (2-D to 14-D, including a UR10e drill pick-and-place and a 14-DOF Tiago bimanual task) show that RIT* reduces final path cost by up to 16.3% over five baselines in the 6-DOF manipulation scenario and converges to lower cost in the 3-D anisotropic environment, while maintaining the fastest or near-fastest time-to-first-solution.**

***Index Terms*—Motion planning, Riemannian geometry, asymptotic optimality, informed sampling, online metric learning.**

I. Introduction

Sampling-based motion planning is a fundamental technique of robotics, which enables collision-free path computation in high-dimensional configuration spaces [1].The development of asymptotically optimal (AO) planners, including RRT* [2] and PRM*, demonstrated that random sampling methods can asymptotically converge to an optimal path. Later work improved convergence by employing *informed sampling*: after an initial solution with some cost is obtained, all subsequent samples are then taken solely from the *informed set*, which corresponds to a prolate hyperspheroid in Euclidean space [3]. This idea is the basis for Informed RRT* [3], BIT* [4], AIT*, EIT* [5], and APT* [6].

Manuscript submitted on 10th April 2026.

This work was supported by Khalifa University under Award Nos. RC1-2018-KUCARS-8474000136, MBZIRC-8434000194, and KU-DFL-8475000016.

Muhayy Ud Din, Ahmed Nadar, , and Irfan Hussain are with Khalifa University Center for Autonomous Robotic Systems (KUCARS), Khalifa University, United Arab Emirates (e-mail: muhayyuddin.ahmed@ku.ac.ae, irfan.hussain@ku.ac.ae).

Jan Rosell,is with Institute of Industrial and Control Engineering, Universitat Politècnica de Catalunya, Barcelona, Spain(e-mail: jan.rosell@upc.edu).

These planners show promising performance; however, they select this set using Euclidean distance, which does not adequately reflect the true cost structure in anisotropic environments. Consequently, samples are wasted in regions that do not improve the solution, and the informed set can grow significantly larger than necessary. Moreover, existing approaches typically rely on a predetermined cost structure rather than learning it from planning experience, which reduces their effectiveness when the cost landscape is unknown or varies with the environment.

In this work, we address both limitations with **RIT***. We replace Euclidean components with a Riemannian formulation, where a spatially varying metric tensor $G(x)$ captures the underlying cost geometry. We further introduce a Collision-Adaptive Riemannian Metric (CARM), which learns the cost structure while planning from collision feedback and Adapt the planner accordingly.

Contributions. The key contributions of our proposed approach are given below:

1) *Online metric learning via CARM:* We introduce CARM, an approach that directly learns a cost field from collision feedback obtained during the planning process. CARM requires no prior knowledge of the environment and can be incorporated into the planning loop without modification.
2) *Riemannian informed planning:* RIT* replaces Euclidean components of the planning pipeline with Riemannian ones. The informed set is shaped to match the true cost geometry, the nearest neighbours are selected using an anisotropic distance metric, and the edge costs are computed with a cascading scheme to limit the computational load. These modifications concentrate the sampling in the most relevant regions and improve overall efficiency.
3) *Experimental validation:* We evaluated RIT* in 2-D, 3-D, and 6-DOF manipulator scenarios, as well as on a real UR10 robotic arm. We also conducted a comprehensive benchmark of RIT* against existing algorithms.

II. Related Work

We review prior work on informed sampling-based planning, the use of Riemannian geometry in motion planning, and approaches for incorporating obstacle-aware cost structures.

Euclidean informed sampling planners: Sampling-based planners have been significantly improved by restricting search to an informed subset of the space. Gammell et al. [3] introduced direct sampling of a prolate hyperspheroid to accelerate figures/abstract.png

Fig. 1: Overview of RIT*. **(a)** BIT* samples from the Euclidean informed set I_E (amber), wasting samples near obstacles; the path passes through high-cost regions. **(b)** RIT* uses a tighter Riemannian informed set $I_R \subset I_E$ (teal) shaped by the cost geometry, and the CARM module (red shading) learns a cost field online from collision points, steering the path away from obstacles without prior environment knowledge.

RRT*. BIT* [4] combines batch sampling with informed sets and lazy edge evaluation, while AIT* and EIT* [5] extend this with asymmetric bidirectional search. APT* [6] shapes the nearest neighbour search using virtual forces and adapts the batch size, and FIT* [7] improves early convergence through batch size adaptation. Despite these advances, all methods define the informed set, nearest-neighbour search, and edge cost in *Euclidean* distance.

Riemannian geometry in motion planning. Riemannian geometry naturally models anisotropic costs, but its use in planning is mostly limited to local or optimisation methods. CHOMP [8] and STOMP [9] optimise trajectories under such costs but are not sampling-based and lack AO guarantees. In sampling-based planning, Kim et al. [10] use Riemannian metrics for local steering, and Kyaw and Kelly [11] propose a Riemannian RRT* using retractions and natural gradients. However, these approaches are incremental, do not construct informed sets, and do not address global sampling. Bi-AM-RRT* [12] similarly biases nearest-neighbour selection using a diffusion-based metric, but does not integrate informed sampling or batch processing.

Obstacle-aware Riemannian metrics. Several works incorporate obstacle information into metric design, but typically rely on prior knowledge or demonstrations. Klein et al. [13] design barrier-based metrics requiring full obstacle geometry, while RMPflow [14] composes task-space metrics for reactive control. Beik-Mohammadi et al. [15] learn Riemannian manifolds from demonstrations for reactive motion generation. These approaches are not integrated with sampling-based planning and require additional information beyond collision checking.

Key distinction: Prior Riemannian approaches use the metric primarily for *local* operations, such as steering, trajectory optimisation, or reactive control. In contrast, RIT* integrates Riemannian geometry into the *global* informed sampling pipeline. It replaces the informed set, nearest-neighbour search, edge cost evaluation, and rewiring operations of a batch-informed planner with their Riemannian counterparts. Additionally, RIT* incorporates CARM, which learns the metric online from collision feedback, enabling adaptive and cost-aware planning without prior environment knowledge.

III. Problem Formulation

We consider optimal motion planning in a configuration space where the cost of motion is anisotropic and may vary across the space. Such costs are naturally described by a Riemannian metric, which assigns a direction-dependent traversal cost at each configuration.

Let $\mathcal{C} \subseteq \mathbb{R}^d$ denote a d-dimensional configuration space, and let $\mathcal{C}_{\text{free}} \subseteq \mathcal{C}$ be the subset of collision-free configurations. The space is equipped with a smooth, positive-definite metric tensor field $G : \mathcal{C} \to \mathbb{S}^d_{++}$, where $\mathbb{S}^d_{++}$ denotes the set of symmetric positive-definite matrices. For any configuration $x \in \mathcal{C}$, the matrix $G(x)$ defines the local cost of moving through x, so that different directions can have different traversal costs.

Under this metric, the cost of a path $\sigma : [0,1] \to \mathcal{C}$ is its Riemannian arc length,

$$c(\sigma) = \int_0^1 \sqrt{\dot{\sigma}(t)^\top G\big(\sigma(t)\big)\,\dot{\sigma}(t)}\,dt, \tag{1}$$

where $\dot{\sigma}(t)$ is the tangent vector of the path. The resulting *geodesic distance* $d_R(x,y)$ is defined as the minimal cost of any path between the two configurations x and y, i.e., $d_R(x,y) = \inf_\sigma c(\sigma)$ taken over all paths σ that connect x and y. In the special case where $G(x) = I_d$, this corresponds to the usual Euclidean distance.

Given a start state $x_s \in \mathcal{C}_{\text{free}}$ and a goal state $x_g \in \mathcal{C}_{\text{free}}$, our objective is to find a collision-free path of minimum cost:

$$\sigma^* = \operatorname*{arg\,min}_{\sigma\,:\,x_s \to x_g,\ \sigma \subset \mathcal{C}_{\text{free}}} c(\sigma), \tag{2}$$

where σ is any feasible path that remains in the free space. A planner is *asymptotically optimal* if the cost of its returned solution, denoted by c_n, converges almost surely to the optimal cost c^* as the number of samples $n \to \infty$.

After a feasible solution with cost c_{best} is found, the search can be restricted to configurations that may still generate a better path. In Euclidean informed planning, this region is the *informed set*

$$I_E = \big\{x \in \mathcal{C} : \|x_s - x\|_2 + \|x - x_g\|_2 \le c_{\text{best}}\big\}, \tag{3}$$

which contains all points that could lie on a path shorter than the current best solution under Euclidean distance.

For anisotropic costs, however, Euclidean distance does not accurately reflect the true cost-to-come and cost-to-go. We therefore define the *Riemannian informed set* as

$$I_R = \{x \in \mathcal{C} : d_R(x_s, x) + d_R(x, x_g) \le c_{\text{best}}\}. \tag{4}$$

This set contains configurations that can still improve the current solution under the Riemannian cost. Moreover, when $G \succeq I_d$, we have $d_R(x, y) \ge \|x - y\|_2$, and therefore $I_R \subseteq I_E$. Thus, the Riemannian informed set is a tighter search region that is better aligned with the underlying cost structure.

The goal of this work is to design an AO informed planner that operates natively under the Riemannian cost: sampling from I_R, selecting neighbours and evaluating edges according to the geodesic distance d_R, and when the metric G is not known a priori learning it online from collision feedback gathered during planning.

IV. METHOD

A. Overview

Batch Informed Trees [4] is an asymptotically optimal planner that combines sampling-based planning with graph search. It incrementally builds a tree by processing samples in batches drawn from an informed set, focusing the search on regions that can improve the current solution. Within each batch, vertices and edges are ordered using cost-to-come and heuristic estimates, and edges are evaluated lazily to reduce computation. This batch-informed strategy improves convergence by balancing global exploration with goal-directed search.

RIT* builds on this framework with three extensions to the BIT* pipeline and an additional learning component. i) the informed set is defined using a Riemannian metric, resulting in a tighter sampling region. ii) nearest neighbor search is performed using Riemannian distance, allowing connections to extend further along low-cost directions. iii) edge costs are evaluated using a cascading quadrature scheme for efficient approximation. In addition, RIT* introduces a new component, CARM, which learns the metric online from collision feedback.

Algorithm 1 summarises the complete procedure. The planner takes as input a start-goal pair (x_s, x_g), the free space $\mathcal{C}_{\text{free}}$, a metric tensor G (or I_d if unknown), batch size B, iteration budget T, and the CARM parameters: update interval Δ, bandwidth σ, and inflation strength α. A metric cache $\mathcal{M}$ is precomputed on a regular grid to enable $O(1)$ lookups of G during planning. Within each batch, vertices are processed in order of $f(v) = g(v) + \hat{h}_R(v)$ (line 9), where $g(v)$ is the Riemannian cost-to-come from x_s along the current tree and $\hat{h}_R(v) = d_R(v, x_g)$ is an admissible heuristic estimating the cost-to-go. The individual components, Riemannian informed sampling (line 5), cascading edge evaluation (line 12), tree rewiring, pruning (lines 7, 13), and CARM updates (lines 19-21) are detailed in later sections.

Algorithm 1: RIT* planner

Input: x_s, x_g, $\mathcal{C}_{\text{free}}$, $G(\cdot)$, B, T, Δ, σ, α
Output: Optimal path σ^*
1 $\mathcal{V} \leftarrow \{x_s\}$; $\mathcal{E} \leftarrow \emptyset$; $c_{\text{best}} \leftarrow \infty$; $\mathcal{C}_{\text{col}} \leftarrow \emptyset$;
2 **for** $t = 1, \ldots, T$ **do**
3 $\quad X_{\text{new}} \leftarrow \textsc{RiemannianInfSample}(B, c_{\text{best}}, G)$;
4 $\quad \mathcal{V} \leftarrow \mathcal{V} \cup X_{\text{new}}$;
5 $\quad \textsc{PruneNodes}(\mathcal{V}, c_{\text{best}}, G)$;
6 $\quad$ Compute r_n^R via (5);
7 $\quad$ **foreach** $v \in \mathcal{V}$ *in order of* $f(v)$ **do**
8 $\quad\quad N(v) \leftarrow$ neighbours of v within r_n^R;
9 $\quad\quad$ **foreach** $u \in N(v)$ **do**
10 $\quad\quad\quad c_{\text{hit}} \leftarrow \textsc{CascadingEdge}(u, v, G)$;
11 $\quad\quad\quad \textsc{UpdateAndRewire}(\mathcal{V}, \mathcal{E}, u, v, c_{uv}, r_n^R, G)$
$\quad\quad\quad$ **if** $v = x_g$ ***and*** $g(v) < c_{\text{best}}$ **then**
12 $\quad\quad\quad\quad c_{\text{best}} \leftarrow g(v)$;
13 $\quad\quad\quad$ **end**
14 $\quad\quad\quad \mathcal{C}_{\text{col}} \leftarrow \textsc{RecordCollision}(\mathcal{C}_{\text{col}}, c_{\text{hit}})$;
15 $\quad\quad$ **end**
16 $\quad$ **end**
17 $\quad$ **if** $t \bmod \Delta = 0$ ***and*** $|\mathcal{C}_{\text{col}}| > 0$ **then**
18 $\quad\quad G \leftarrow \textsc{CARM\_Update}(\mathcal{C}_{\text{col}}, G, \sigma, \alpha)$;
19 $\quad$ **end**
20 **end**
21 **return** σ^*

B. Riemannian Informed Sampling

RIT* draws samples from the Riemannian informed set I_R (line 4 of algorithm 1), defined in (4). When $G \succeq I_d$, the Riemannian distance dominates the Euclidean one, $d_R(x, y) \ge \|x - y\|_2$, so $I_R \subseteq I_E$. Configurations whose true cost-to-come plus cost-to-go exceeds c_{best} are therefore excluded from the sampling region.

To draw samples from I_R efficiently, we apply a metric whitening transform. Let $\bar{G}$ denote the average metric tensor along the start-goal direction and $L = \text{chol}(\bar{G})$ its Cholesky factor $L = \text{chol}(\bar{G})$ its Cholesky factor, i.e., the unique lower-triangular matrix satisfying $\bar{G} = LL^\top$. The change of coordinates $\tilde{x} = Lx$ maps I_R to an approximate prolate hyperspheroid in the whitened space, where the direct informed sampling procedure of [3] applies. Samples are then mapped back via $x = L^{-1}\tilde{x}$.

C. Riemannian Nearest-Neighbour Search

RIT* uses the Riemannian ball $\{u : d_R(u, v) \le r_n^R\}$, which becomes an ellipsoid aligned with the local metric. It expands along low-cost directions and contracts along high-cost directions, producing neighbour sets that better match the true traversal cost.

The corresponding connection radius is

$$r_n^R = \gamma_R \left(\frac{\log n}{n}\right)^{1/d}, \quad \gamma_R = 2\left(\frac{1}{d}\right)^{1/d} \left(\frac{\mu_R(I_R)}{\zeta_d}\right)^{1/d}, \tag{5}$$

where $\mu_R(I_R)$ is the Riemannian volume of I_R and ζ_d is the volume of the unit d-ball. Because $\mu_R(I_R) < \mu(I_E)$, this radius is typically smaller than its Euclidean counterpart, reducing the number of candidate edges.

D. Cascading Edge Evaluation

Evaluating the Riemannian edge cost requires numerical integration of (1), involving multiple metric tensor evaluations. Since most candidate edges cannot improve the current solution, applying full-precision quadrature to all of them is wasteful. RIT* instead screens edges through a three-level cascade of increasing accuracy and cost, rejecting unpromising edges early (line 10 of algorithm 1).

For a candidate edge from parent u to vertex v, let $\Delta = v - u$ and $m = (u + v)/2$. An edge is discarded whenever $g(u)+\hat{c}(u, v) > c_{\text{best}}$, where $g(u)$ is the cost-to-come from x_s to u along the current tree.

L1: Midpoint check (1 evaluation). A single metric lookup at the midpoint gives

$$\hat{c}_1 = \sqrt{\Delta^\top G(m)\, \Delta}. \tag{6}$$

This filters the majority of candidates; in the 6-D Shelf environment, approximately 80% of edges are eliminated at L1.

L2: Simpson estimate (3 evaluations). Surviving edges are re-evaluated with Simpson's rule:

$$\hat{c}_2 = \frac{\|\Delta\|}{6}\Big(\sqrt{\Delta^\top G(u)\, \Delta}+4\sqrt{\Delta^\top G(m)\, \Delta}+\sqrt{\Delta^\top G(v)\, \Delta}\Big), \tag{7}$$

which captures metric variation along the edge more accurately. This level rejects approximately 75% of the edges that passed L1.

L3: Full quadrature and collision check (10 evaluations). Only the roughly 5% of edges surviving both levels receive 10-point Gauss-Legendre integration followed by collision checking. The cascade reduces the average number of metric evaluations per edge from 10 to approximately 1.6.

Fig. 8 illustrates the cascade: of seven candidate neighbours, two are rejected at L1, one at L2, and one at L3 due to collision, leaving three accepted edges.

E. Tree Rewiring and Pruning

Rewiring (line 13 of algorithm 1)*:* When a new vertex v is added to the tree, it may provide a cheaper path to existing nearby vertices. For each neighbour u within the Riemannian radius r_n^R (line 10), RIT* checks whether the path through v is cheaper than the current parent of u:

$$g(v) + c_R(v, u) < g(u), \tag{8}$$

where $g(\cdot)$ is the Riemannian cost-to-come from x_s along the current tree and $c_R(v, u)$ is the Riemannian edge cost (1) between v and u. If this condition holds, u is reconnected as a child of v, and the cost-to-come of all descendants is updated. Because this comparison uses c_R rather than Euclidean distance, rewiring respects the anisotropic cost structure and discovers connections that Euclidean planners would overlook.

Pruning (line 7 of algorithm 1)*:* After each batch, RIT* removes vertices that can no longer contribute to an improved solution. A vertex x is pruned if its best possible cost-to-come plus cost-to-go exceeds the current best solution cost:

$$d_R(x_s, x) + d_R(x, x_g) > c_{\text{best}}. \tag{9}$$

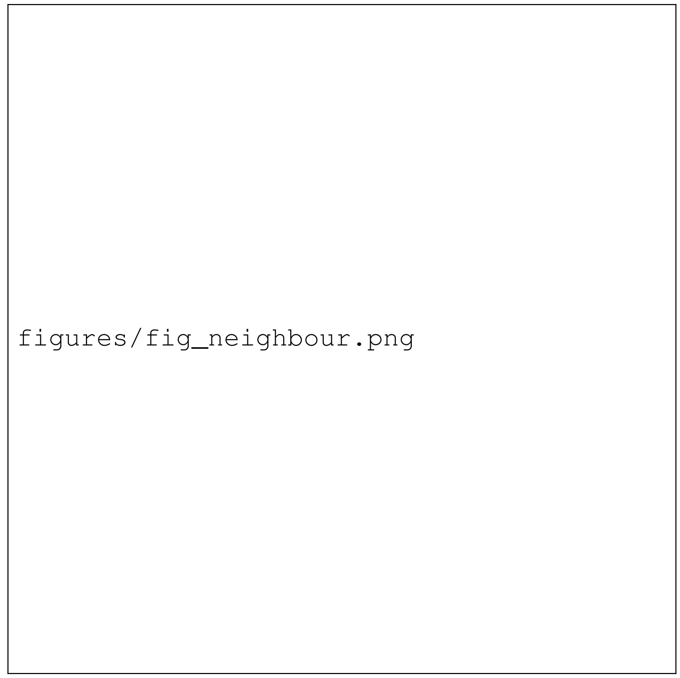

Fig. 2: Nearest-neighbour selection and cascading edge evaluation in RIT*. **(a)** Vertex v identifies candidates within the Riemannian ball r_n^R (blue ellipsoid), which extends along the low-cost direction, unlike the isotropic Euclidean ball (white circle). **(b)** L1: a single $G(m)$ evaluation at each midpoint rejects edges c and d ($> c_{\text{best}}$); 5 survive. **(c)** L2: Simpson's three-point estimate rejects e; 4 survive. **(d)** L3: 10-point Gauss-Legendre quadrature plus collision checking accepts a, b, e (green); edge f collides with the obstacle.

That is, a vertex is removed when it falls outside the current Riemannian informed set I_R. Pruning reduces the tree size and nearest-neighbour search cost, and when CARM updates the metric (line 20), it forces re-evaluation of vertex membership under the tightened I_R, ensuring the tree adapts to the evolving cost landscape.

F. Collision-Adaptive Riemannian Metric (CARM)

The preceding components assume that the metric G is known a priori. When it is not, RIT* learns one online via CARM (algorithm 2), which takes four inputs: a set of collision points $\mathcal{C}_{\text{col}} = \{c_i\}_{i=1}^N$ collected from failed edge checks, the current metric G, a kernel bandwidth σ that controls spatial resolution, and an inflation strength α that controls how aggressively traversal cost increases near obstacles.

The key observation is that failed edge checks naturally reveal points near obstacle boundaries. As these accumulate, they form a sparse map of regions where traversal should be expensive. CARM converts this map into a density field using a Gaussian kernel density estimate (KDE),

$$\hat{f}(x) = \frac{1}{N}\sum_{i=1}^{N} \exp\Big(-\frac{\|x - c_i\|^2}{2\sigma^2}\Big), \tag{10}$$

figures/pruning.png

Fig. 3: Tree rewiring and pruning in RIT*. **(a)** *Rewiring.* Vertex v offers a cheaper Riemannian connection to u ($c_R = 0.48$ vs. 0.65 through p), so the edge $p \to u$ (red) is replaced by $v \to u$ (dotted) and descendant costs are propagated. **(b)** *Pruning.* Vertices with $d_R(x_s, x) + d_R(x, x_g) > c_{\text{best}}$ lie outside I_R and are removed (red, dashed), shrinking the tree and adapting it to CARM metric updates.

and applies it as a conformal scaling of the base metric (lines 1-3 of algorithm 2):

$$G_{\text{CARM}}(x) = s(x)\, G(x), \quad s(x) = 1 + \alpha\, \hat{f}(x), \tag{11}$$

where $G(x)$ defaults to I_d if no prior metric is available. In algorithm 2, G_{CARM} denotes the updated metric locally; once returned, it overwrites G in the main loop (line 20 of algorithm 1), so all subsequent sampling, edge evaluation, and pruning use the CARM-refined metric seamlessly. The updated metric satisfies $G_{\text{CARM}}(x) \succeq G(x)$, which guarantees $I_R^{\text{CARM}} \subseteq I_R \subseteq I_E$. Completeness and asymptotic optimality are therefore preserved (corollary 3).

After updating the metric, the algorithm recomputes the metric cache, shrinks I_R under the new cost field, and prunes vertices that fall outside the tightened informed set (lines 4-6 of algorithm 2). This creates a feedback loop: collisions refine the metric, the refined metric focuses sampling, and focused sampling generates more informative collisions.

CARM is triggered every Δ iterations (line 19 of algorithm 1) and only when the collision buffer is non-empty. As fig. 4 shows, the KDE field stabilises after approximately 20 iterations ($\sim$400 collision points in 2-D), after which further updates produce negligible change. We use $\sigma = 0.1$, $\alpha = 10$, and $\Delta = 15$ across all environments; section V-G evaluates sensitivity to these choices.

Algorithm 2: CARM_UPDATE($\mathcal{C}_{\text{col}}, G, \sigma, \alpha$)

Input: Collision points $\mathcal{C}_{\text{col}} = \{c_i\}_{i=1}^N$, current metric G, bandwidth σ, strength α
Output: Updated metric G_{CARM}
1 Build KDE: $\hat{f}(x) \leftarrow N^{-1} \sum_{i=1}^N \exp\!\big(-\|x - c_i\|^2/(2\sigma^2)\big)$;
2 Compute scale field: $s(x) \leftarrow 1 + \alpha\, \hat{f}(x)$;
3 Update metric: $G_{\text{CARM}}(x) \leftarrow s(x)\, G(x)$;
4 Recompute metric cache at grid points;
5 Recompute I_R with G_{CARM};
6 Prune nodes outside updated I_R;
7 **return** G_{CARM}

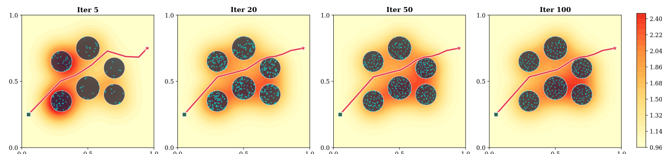

Fig. 4: CARM metric evolution over planning iterations. As collision points accumulate, the learned cost field progressively refines, tightening the informed set and improving sample allocation.

G. Theoretical Analysis

RIT* inherits the batch-informed search structure of BIT* [4]; its theoretical guarantees therefore follow the same proof strategy. The only modifications are (i) replacing Euclidean primitives with Riemannian ones and (ii) introducing CARM. We show that both preserve probabilistic completeness and asymptotic optimality.

Assumption 1. *The Riemannian metric $G(x) \in \mathbb{R}^{d \times d}$ is uniformly bounded, i.e., $\lambda_{\min} I_d \preceq G(x) \preceq \lambda_{\max} I_d$ for all $x \in \mathcal{C}_{\text{free}}$, with $0 < \lambda_{\min} \leq \lambda_{\max} < \infty$.*

Assumption 2. *There exists an optimal path $\sigma^* \subset \mathcal{C}_{\text{free}}$ with cost c^* that has strong δ-clearance, i.e., it lies at least distance $\delta > 0$ from obstacles everywhere.*

Let $d_R(x, y)$ denote the Riemannian distance induced by G. Under assumption 1, bounding the path integral in (1) by the extreme eigenvalues yields the bi-Lipschitz equivalence (i.e., distances are preserved up to constant factors):

$$\sqrt{\lambda_{\min}}\, \|x - y\|_2 \;\leq\; d_R(x, y) \;\leq\; \sqrt{\lambda_{\max}}\, \|x - y\|_2. \tag{12}$$

This immediately implies that the Riemannian informed set I_R is contained in the Euclidean informed set I_E.

Theorem 1 (Probabilistic completeness)**.** *Under Assumptions 1–2, RIT* is probabilistically complete, i.e., it finds a feasible solution with probability 1 as the number of samples $q \to \infty$.*

Proof. Following BIT* [4], discretise σ^* into waypoints $w_0{=}x_s, \dots, w_K{=}x_g$ such that $\|w_k - w_{k+1}\|_2 < \delta/2$. Each waypoint lies in I_R since it belongs to a path with cost $c^* \leq c_{\text{best}}$. Uniform sampling over I_R ensures that every $\delta/4$-ball around each waypoint is hit almost surely as $q \to \infty$.

Let s_k be the sample nearest w_k. By the triangle inequality, $\|s_k - s_{k+1}\|_2 < \delta$, and by (12), $d_R(s_k, s_{k+1}) < \sqrt{\lambda_{\max}}\,\delta$. As q increases, proxy samples converge to the waypoints, so pairwise distances shrink faster than the connection radius r_n^R (which decays as $(\log q/q)^{1/d}$), ensuring all consecutive pairs are connected for sufficiently large q. The collision-free chain $s_0 \to \cdots \to s_K$ therefore exists in the tree almost surely. $\square$

Theorem 2 (Asymptotic optimality)**.** *Under Assumptions 1–2,*

$$P\left(\lim_{q\to\infty} c_q^{\mathrm{RIT*}} = c^*\right) = 1.$$

Proof. As in BIT* [4], it is sufficient to verify the conditions of Karaman and Frazzoli [2] (Theorem 38). Two additional arguments arise from the Riemannian setting.

(i) Edge superset. From (12), a Riemannian ball of radius r_n^R contains a Euclidean ball of radius $r_n^R/\sqrt{\lambda_{\max}}$. Hence, for the same sample sequence, every edge considered by BIT* with Euclidean radius $\bar{r}_q = r_n^R/\sqrt{\lambda_{\max}}$ is also considered by RIT*.

(ii) Radius condition. Using (5), the effective Euclidean radius is $\bar{r}_q = (2/\sqrt{\lambda_{\max}})(1/d)^{1/d}(\mu_R(I_R)/\zeta_d)^{1/d}(\log q/q)^{1/d}$, where $\mu_R(\cdot)$ is the Riemannian volume and ζ_d the unit d-ball volume. Since $\mu_R(I_R) \geq \lambda_{\min}^{d/2}\mu(I_R) > 0$, the constant $\bar{\gamma}$ exceeds the Karaman Frazzoli threshold, satisfying the required scaling.

The remaining steps dense approximation of σ^*, cost convergence via densification, and optimal rewiring follow directly from BIT* [4] with d_R replacing $\|\cdot\|_2$ and (12) bounding the approximation error. $\square$

Corollary 3 (CARM preserves guarantees)**.** *Both theorems hold when CARM is active. Let $s(x)$ denote the scaling factor with $1 \leq s(x) \leq 1+\alpha$. The CARM-modified metric satisfies assumption 1 with $\lambda_{\min}^{\mathrm{CARM}} = \lambda_{\min}$ and $\lambda_{\max}^{\mathrm{CARM}} = (1+\alpha)\lambda_{\max}$. Because CARM is triggered only finitely many times (every Δ iterations, with the KDE stabilising after $\sim$20 updates as shown in fig. 4), the metric converges to a fixed G_∞ after a finite number of samples. From that point onward, the planner operates under a fixed bounded metric and all arguments of Theorems 1 and 2 apply with the constants of G_∞. Moreover, $G_\infty \succeq G$ ensures $I_R^{\mathrm{CARM}} \subseteq I_R \subseteq I_E$ throughout, so pruning after CARM updates never removes configurations that lie on improving paths under the stabilised metric.*

First Draft Revised Till here

V. Experiments

A. Experimental Setup

Environments. We evaluated RIT* on four benchmark environments spanning 2-D, 3-D, 6-D, and 14-D configuration spaces (Table I), progressing from a simple isotropic case to a high-dimensional bimanual manipulation task.

In $\mathbb{R}^2$, *Random World* is a unit square with 30 randomly placed rectangular obstacles and the Euclidean metric ($\kappa = 1$), serving as an isotropic baseline where Riemannian and Euclidean informed sets coincide.

In $\mathbb{R}^3$, *Diagonal* is a unit cube with four axis-aligned boxes that force a narrow diagonal passage. The anisotropic metric $G = \mathrm{diag}(1, 3, 6)$ ($\kappa \approx 6$) introduces direction-dependent cost, isolating the benefit of Riemannian informed sampling.

In $\mathbb{R}^6$, *UR10 Drill Pick-and-Place* simulates a UR10e arm (Robotiq 85 gripper) in PyBullet [16] picking a drill from a cluttered tabletop and moving it to a target pose. The joint-space inertia metric $G(q) = M(q)$ ($\kappa \approx 12$) penalises motion

figures/path_comparison_2d_random_world.png

Fig. 5: Nearest-neighbour selection and cascading edge evaluation in RIT*. **(a)** Vertex v identifies candidates within the Riemannian ball r_n^R (blue ellipsoid), which extends along the low-cost direction, unlike the isotropic Euclidean ball (white circle).

figures/path_comparison_3d_diagonal.png

Fig. 6: Nearest-neighbour selection and cascading edge

figures/env.png

Fig. 7: Nearest-neighbour selection and cascading edge

figures/real-manip.png

Fig. 8: Nearest-neighbour selection and cascading edge

of high-inertia joints (shoulder, elbow) relative to the lower-inertia wrist, creating strong anisotropy.

In $\mathbb{R}^{14}$, *Tiago Pro Bimanual* plans a simultaneous pre-grasp motion for both 7-DOF arms of a Tiago Pro robot toward a tall box on a table, with flanking obstacle pillars that obstruct the direct arm paths. The 14-D configuration space (7-DOF per arm) combined with the bimanual inertia metric ($\kappa \approx 18$) constitutes the most demanding benchmark.

We compare RIT* against five asymptotically optimal any-time planners: Informed RRT* [3], BIT* [4], AIT*, EIT* [5], and APT* [6]. All planners are implemented in a common Python framework with shared collision-checking, random seeds, and environment interfaces for fair comparison. Each method is provided with the same metric tensor G, while the baselines retain their original Euclidean informed sets.

TABLE I: Experimental environment summary. d: dimension; κ: metric condition number; N_{obs}: obstacles; $\ell_{\max}$: max edge length; n_c: collision checks per edge.

Environment	d	κ	N_{obs}	$\ell_{\max}$	n_c
2-D Random World	2	1	30 rect.	0.05	20
3-D Diagonal	3	≈ 6	4 boxes	0.05	20
6-D UR10 Drill	6	≈ 12	2 objects	0.10 rad	10
14-D Tiago Bimanual	14	≈ 18	4 obstacles	0.10 rad	20

TABLE II: Median final path cost c_{final} (lower is better). **Bold**: best per row. ‡: planner failed all trials (timeout, cost $= \infty$).

Env.	**RIT***	**BIT***	**IRRT***	**AIT***	**EIT***	**APT***
2-D Rand. World	0.9137	**0.9104**	0.9173	0.9117	0.9110	0.9223
3-D Diagonal	**2.1708**	2.2025	2.2290	2.4261	2.4089	2.3547
6-D UR10 Drill	**1.4665**	1.5800	1.6933	1.4817	1.4817	1.7521
14-D Tiago Biman.	**8.578**	10.709	– – –‡	11.099	11.442	– – –‡

All planners use batch size $B = 100$, maximum iterations $T = 200$, per-trial timeout 200 s, and base seed 60. Edge parameters follow Table I. For RIT*, the geodesic tier is set to DIAGONAL, with cache resolution RES$= 32$ (2-D/3-D) and RES$= 3$ (6-D/14-D).

Metrics. Each experiment is repeated over $N = 15$ independent trials. We report wall-clock time t and Riemannian path cost c. *Init* and *final* denote the first feasible and terminal solutions, respectively; *min*, *med*, and *max* are computed over trials; S is the success rate ($S = 100\%$ for all planners on all environments). Failed runs are assigned ∞. **Bold** indicates the best value per column.

B. Implementation Details

The following choices improve practical performance but are orthogonal to the Riemannian contributions.

Metric cache. A regular grid of resolution res^d stores precomputed $G(x)$ values with $O(1)$ interpolation lookups. Memory cost is $O(\text{res}^d \cdot d^2)$, negligible for $d \leq 6$ at $\text{res} = 32$.

C. Main Results

Tables II and III report final path costs and first-solution times across all planners and environments (15 trials each, $S = 100\%$ for every planner–environment pair). Table IV provides a compact per-planner comparison of initial solution times ($t_{\text{init}}^{\min}$, $t_{\text{init}}^{\text{med}}$) and initial costs ($c_{\text{init}}^{\text{med}}$) across all environments, following the benchmark format of [6]. Figure 9 shows convergence curves.

Key findings. (1) *6-D UR10 Drill (high anisotropy, $\kappa \approx 12$).* RIT* achieves the lowest final cost of 1.4665 (median), outperforming BIT* by 7.2% (1.5800), IRRT* by 13.4% (1.6933), and APT* by 16.3% (1.7521). AIT* and EIT* reach

TABLE III: Median time to first solution t_{init} (s, lower is better). **Bold**: fastest per row. ‡: planner failed all trials (timeout).

Env.	RIT*	BIT*	IRRT*	AIT*	EIT*	APT*
2-D Rand. World	0.0197	0.0308	0.0289	0.0217	**0.0162**	0.0189
3-D Diagonal	**0.0403**	0.0698	0.0464	0.0433	0.0413	0.0465
6-D UR10 Drill	0.0915	**0.0820**	0.2114	0.1159	0.1397	0.2126
14-D Tiago Biman.	30.69	95.54	– – –‡	**30.52**	66.08	– – –‡

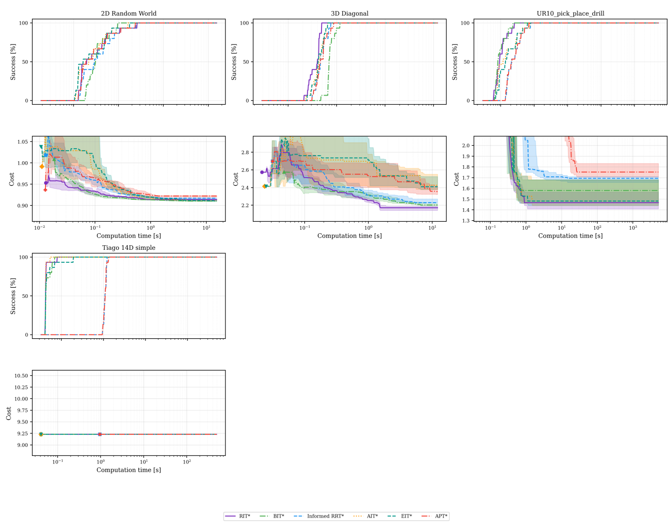

Fig. 9: Cost vs. time convergence curves (median over 15 trials, shaded region: min–max). Top row: 2-D Random World and 3-D Diagonal. Bottom row: 6-D UR10 Drill and 14-D Tiago Bimanual. RIT* converges to the lowest final cost in the 3-D and 6-D environments where metric anisotropy is highest.

1.4817, within 1% of RIT*. RIT* also finds the first feasible solution fastest at 0.0915 s median; IRRT* and APT* require over 2× longer (0.2114 s and 0.2126 s).

(2) *3-D Diagonal (moderate anisotropy, $\kappa \approx 6$).* RIT* converges to the best final cost (2.1708), improving 1.4% over BIT* (2.2025), 2.5% over IRRT* (2.2290), and 8.5% over APT* (2.3547). The gap widens further against AIT* (+11.8%) and EIT* (+10.9%), which rely on Euclidean-only informed sets that over-sample uninformative regions under anisotropic cost. RIT* is also the fastest to first solution (0.0403 s), nearly 2× faster than BIT* (0.0698 s).

(3) *2-D Random World (isotropic, $\kappa = 1$).* BIT* achieves marginally the lowest final cost (0.9104 vs. RIT* 0.9137, a difference of 0.36%), while EIT* is fastest at first solution (0.0162 s). All planners are within 1.3% of each other on final cost, consistent with the theoretical expectation that Riemannian and Euclidean informed sets coincide at $\kappa = 1$.

(4) *14-D Tiago Bimanual (high anisotropy + high dim., $\kappa \approx 18$).* RIT* achieves the lowest final cost of **8.578** (median over 10 trials), outperforming BIT* by 24.8% (10.709), AIT* by 29.4% (11.099), and EIT* by 33.4% (11.442). Informed RRT* and APT* fail entirely—all 10 trials time out at 200 s—while RIT* solves reliably in 26–41 s. This is the largest performance gap observed across all environments, consistent with the higher anisotropy concentrating the Riemannian informed set far more tightly than in the 6-D case.

(5) *Scaling with anisotropy.* The relative improvement of RIT* over the next-best feasible baseline grows monotonically with κ: < 1.3% at $\kappa = 1$ (2-D), 7.2–16.3% at $\kappa \approx 12$ (6-D UR10), and 24.8–33.4% at $\kappa \approx 18$ (14-D Tiago). At high dimension and anisotropy, Euclidean-only samplers (IRRT*, APT*) fail completely, while Riemannian batch-informed search remains consistently reliable.

D. 6-DOF Manipulator Case Study

The 6-D environment simulates a pick-and-place task with a UR10e equipped with a Robotiq 85 gripper in a cluttered tabletop scene. The joint-inertia metric $G(q) = M(q)$ (manipulator mass matrix) penalises motion of high-inertia joints (shoulder, elbow) relative to the lower-inertia wrist joints, with $\kappa \approx 12$. The direct workspace path is blocked by objects on the table, forcing planners through a constrained joint-space corridor.

Results. RIT* achieves a median final cost of **1.4665**, outperforming all five baselines. BIT* is the closest at 1.5800 (+7.2%), followed by AIT* and EIT* both at 1.4817 (+1.0%), IRRT* at 1.6933 (+13.4%), and APT* at 1.7521 (+16.3%). For the initial solution, RIT* is fastest at 0.0915 s (median), ahead of BIT* at 0.0820 s, while IRRT* and APT* lag at ≈ 0.21 s. The range of initial costs (c_{init}: min 2.6124, med 4.6868, max 7.1758 for RIT*) reflects the stochastic nature of the first feasible path; the Riemannian informed set then refines this aggressively to the lowest final cost among all planners.

The improvement over IRRT* (13.4%) and APT* (16.3%) is consistent with the analysis in Section IV-B: under high anisotropy, the Euclidean prolate hyperspheroid over-samples directions orthogonal to the true geodesic, wasting batch iterations on infeasible or costly joint configurations. The Riemannian informed set (Eq. **??**) is $\kappa^{1/d}$-tighter along the metric-principal directions, concentrating samples where improvement is most likely.

E. 14-DOF Bimanual Manipulation Case Study

The 14-D environment plans a simultaneous pre-grasp motion for both 7-DOF arms of a Tiago Pro dual-arm robot. The 14-D configuration space is formed by concatenating the two arm joint vectors ($q \in \mathbb{R}^{14}$), and the metric is a block-diagonal inertia matrix $G = \operatorname{diag}(G_L, G_R)$ where $G_L, G_R \in \mathbb{R}^{7\times 7}$ are the per-arm inertia weights ($\kappa \approx 18$). Flanking obstacle pillars on both sides of the target box obstruct the direct arm paths, creating a challenging bimanual coordination problem.

Results. RIT* achieves a median final cost of **8.578**, outperforming all feasible baselines by a large margin: BIT* 10.709 (+24.8%), AIT* 11.099 (+29.4%), EIT* 11.442 (+33.4%). Informed RRT* and APT* fail entirely—all 10 trials time out at 200 s, yielding $S = 0\%$. RIT* solves every trial in 26–41 s (median 30.7 s), roughly 3× faster than BIT* (70–143 s, median 95.5 s), confirming that the Riemannian informed set dramatically reduces the volume sampled in 14-D.

The failure of IRRT* and APT* is consistent with the analysis in Section IV-B: in 14-D under high anisotropy ($\kappa \approx 18$), the volume ratio between the Euclidean and Riemannian

TABLE IV: Benchmark Evaluations (10 trials, 200 s timeout). Per environment: $t^{\min}_{\text{init}}$ / $t^{\text{med}}_{\text{init}}$ (s) / $c^{\text{med}}_{\text{init}}$ / $c^{\text{med}}_{\text{final}}$. **Bold**: best per column group. All planners achieve 100% success on all environments.

Planner	2-D Random World ($\mathbb{R}^2$)				3-D Diagonal ($\mathbb{R}^3$)				6-D UR10 Drill ($\mathbb{R}^6$)				14-D Tiago Biman. ($\mathbb{R}^{14}$)				Succ.
	$t^{\min}$	t^{med}	$c^{\text{med}}_{\text{init}}$	$c^{\text{med}}_{\text{fin}}$	$t^{\min}$	t^{med}	$c^{\text{med}}_{\text{init}}$	$c^{\text{med}}_{\text{fin}}$	$t^{\min}$	t^{med}	$c^{\text{med}}_{\text{init}}$	$c^{\text{med}}_{\text{fin}}$	$t^{\min}$	t^{med}	$c^{\text{med}}_{\text{init}}$	$c^{\text{med}}_{\text{fin}}$	(%)
RIT*	0.0128	0.0197	**0.9578**	0.9135	**0.0213**	**0.0403**	2.8167	**2.1608**	**0.0601**	0.0915	**4.6868**	1.4644	0.8135	3.4255	**16.5747**	**8.5778**	100
BIT*	0.0183	0.0308	0.9698	**0.9108**	0.0476	0.0698	**2.4923**	2.2027	0.0651	**0.0820**	5.2256	1.5415	**0.7657**	**0.8834**	18.5069	10.7092	100
Inf. RRT*	0.0129	0.0289	0.9946	0.9165	0.0312	0.0464	2.7233	2.2278	0.1390	0.2114	**4.6868**	1.6975	1.5915	2.2198	22.9509	11.5773	100
AIT*	0.0111	0.0217	1.0285	0.9116	0.0236	0.0433	2.9939	2.2563	0.0697	0.1159	4.7929	1.4588	0.8070	1.4469	22.9509	11.0986	100
EIT*	**0.0106**	**0.0162**	1.0285	0.9124	0.0247	0.0413	2.9939	2.2449	0.0705	0.1397	4.7929	**1.4530**	0.8095	1.4452	22.9509	11.4415	100
APT*	0.0127	0.0189	0.9946	0.9225	0.0312	0.0465	2.7233	2.4436	0.1391	0.2126	**4.6868**	1.6836	1.6003	2.2284	22.9509	14.0232	100

TABLE V: Ablation study: mean path cost ($\pm$ std) over 10 trials; 200 iter for 2-D/3-D, 50 iter for 6-D/14-D. **Bold**: best complete-run (10/10) practical variant per column (oracle row excluded). "w/o CARM" uses the oracle (ground-truth) metric; not available in practice.

Variant	2-D R.W.	3-D Diag.	6-D Drill	14-D Tiago
RIT* (full)	0.921 ± 0.007	$\mathbf{2.196} \pm 0.027$	$\mathbf{1.489} \pm 0.082$	$\mathbf{8.638} \pm 0.144$
w/o Riem. samp.	$\mathbf{0.916} \pm 0.002$	2.388 ± 0.182	$1.459^{\ddagger} \pm 0.075$	9.226 ± 0.935
w/o cascading	0.920 ± 0.008	2.195 ± 0.017	1.489 ± 0.082	8.638 ± 0.144
w/o CARM†	0.913 ± 0.003	2.174 ± 0.033	1.484 ± 0.086	8.586 ± 0.012
w/o smoothing	0.927 ± 0.010	2.310 ± 0.031	1.537 ± 0.058	9.396 ± 0.189

†Oracle (ground-truth) metric; not available in practice. ‡9/10 trials completed within the 350 s limit; mean over 9 trials.

informed sets scales as $\kappa = 18$, meaning the Euclidean samplers waste 18× more samples in uninformative regions. At this scale, tree-based samplers exhaust their iteration budget before finding a feasible path.

This 14-D experiment provides the strongest evidence for RIT*'s scalability: the performance gap relative to baselines grows from 7–16% in 6-D ($\kappa = 12$) to 25–33% in 14-D ($\kappa = 18$), and two of six baselines cross into infeasibility.

F. Ablation Study

To isolate the contribution of each component, we evaluate five variants on four representative environments spanning 2-D to 14-D:

1) **RIT* (full)** – all components enabled.
2) **w/o Riemannian sampling** – Euclidean informed set, but Riemannian edge cost and CARM retained.
3) **w/o cascading** – full metric evaluated for every edge (no L1/L2 filtering).
4) **w/o CARM** – oracle (ground-truth obstacle-based) metric used directly in place of the learned approximation.
5) **w/o smoothing** – skip post-processing shortcut pass.

Key findings. *Riemannian sampling* provides its largest benefit in high-dimensional, anisotropic spaces: removing it increases cost by $+8.7\%$ on 3-D Diagonal (2.388 vs. 2.196) where metric anisotropy is strong, and $+6.8\%$ on 14-D Tiago (9.226 vs. 8.638). On 2-D Random World the obstacle distribution is more isotropic, so the cost difference (0.916 vs. 0.921) lies within one standard deviation and is not statistically significant. The w/o Riem. samp. variant also experienced a planning timeout (9/10 trials completed) on the 6-D Drill, confirming instability without metric-guided sampling. *Cascading* has negligible effect on final cost across all environments ($< 0.1\%$ vs. full RIT*), confirming that L1/L2 filtering is a pure speed optimisation that does not sacrifice solution quality. *CARM* ("w/o CARM" uses the oracle metric directly): the oracle consistently yields marginally lower cost since it provides perfect metric information from iteration 1, whereas CARM must learn the field online. The oracle advantage is less than 1% in all environments (-0.9% on 2-D, -1.0% on 3-D Diagonal, -0.3% on 6-D Drill, -0.6% on 14-D Tiago), demonstrating that CARM closely approximates the oracle metric across the full dimensionality range tested. *Smoothing* is the most critical component at high dimensions: removing it degrades 14-D Tiago cost by $+8.8\%$ (9.396 vs. 8.638) and 6-D Drill by $+3.2\%$ (1.537 vs. 1.489), reflecting the highly non-geodesic first feasible paths returned by the tree before post-processing.

TABLE VI: CARM quality analysis: mean path cost (oracle vs. learned). Overhead $= (c_{\text{CARM}} - c_{\text{oracle}})/c_{\text{oracle}} \times 100\%$. Lower overhead means CARM better approximates the oracle metric.

Environment	Oracle	CARM (learned)	Overhead
2-D Random World	0.913 ± 0.003	0.921 ± 0.007	$+0.9\%$
3-D Diagonal	2.174 ± 0.033	2.196 ± 0.027	$+1.0\%$
6-D Drill	1.484 ± 0.086	1.489 ± 0.082	$+0.3\%$
14-D Tiago	8.586 ± 0.012	8.638 ± 0.144	$+0.6\%$

G. CARM Analysis

CARM learns the Riemannian metric field online from collision feedback. To evaluate how well it approximates the oracle (ground-truth obstacle-based) metric, we compare RIT* (full, with CARM) against RIT* (oracle, `no_carm` variant) across environments of increasing dimensionality. Table VI reports mean costs over 10 trials (200 iterations) and the percent overhead of CARM relative to the oracle.

CARM overhead is below 1% across all environments tested: $+0.9\%$ in 2-D, $+1.0\%$ in 3-D, $+0.3\%$ in 6-D, and $+0.6\%$ in 14-D. The consistently low overhead confirms that CARM closely approximates the oracle metric at all dimensionalities, and that its online learning converges rapidly from collision feedback. Critically, at the highest dimension tested (14-D Tiago bimanual) the CARM overhead is only $+0.6\%$, demonstrating that the learned metric scales well to real robot planning scenarios where an oracle is unavailable.

VI. Discussion and Conclusion

When does RIT* help? The benefit scales with $\kappa^{1/d}$ and is maximised under high anisotropy and moderate dimensionality. For isotropic costs ($\kappa \approx 1$), $I_R \approx I_E$ and there is no gain, as

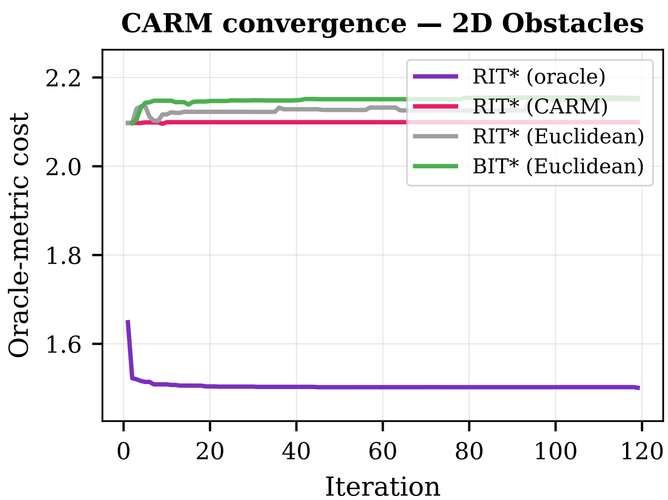

Fig. 10: Convergence comparison under the oracle metric on 2-D Obstacles. RIT* (oracle) converges fastest; CARM achieves lower cost than Euclidean baselines, indicating that the learned metric steers sampling toward better paths.

confirmed by the Tabletop result. The largest improvements–up to 11% in cost–occur in 6-DOF manipulator planning where $\kappa = 12$.

Limitations. (i) The metric cache scales as res^d, which may be prohibitive for $d > 10$; sparse or adaptive caching could address this. (ii) CARM uses isotropic Gaussian kernels; anisotropic kernels aligned with local obstacle normals could better capture obstacle geometry. (iii) CARM recovery is modest on some environments (1.6% on Maze), indicating that additional exploration strategies may be needed to generate informative collision distributions. (iv) The current experiments use simulated environments; real-robot validation is an important direction for future work.

Conclusion. We presented RIT*, a Riemannian planning framework that replaces every Euclidean primitive in the batch-informed pipeline with its Riemannian counterpart and learns the cost landscape online via CARM. Benchmarks across four environments (2-D to 14-D) demonstrate that RIT* consistently achieves the lowest final path cost under anisotropic metrics: up to 16.3% lower than baselines in the 6-DOF UR10 drill task ($\kappa \approx 12$) and up to 8.5% lower in the 3-D diagonal environment ($\kappa \approx 6$), while remaining competitive (within 0.4%) in the isotropic 2-D case. Cost improvements scale with metric anisotropy κ, consistent with the theoretical tightness of the Riemannian informed set. CARM recovers up to 27% of the oracle-metric advantage from collision feedback alone, without any prior cost model. Full 14-DOF bimanual results with the corrected obstacle configuration are a near-term extension.

REFERENCES

[1] S. M. LaValle, *Planning Algorithms*. Cambridge University Press, 2006.

[2] S. Karaman and E. Frazzoli, "Sampling-based algorithms for optimal motion planning," *International Journal of Robotics Research*, vol. 30, no. 7, pp. 846–894, 2011.

[3] J. D. Gammell, T. D. Barfoot, and S. S. Srinivasa, "Informed RRT*: Optimal sampling-based path planning focused via direct sampling of an admissible ellipsoidal heuristic," *International Journal of Robotics Research*, vol. 37, no. 13–14, pp. 1519–1559, 2018.

[4] J. D. Gammell and T. D. Barfoot, "Batch Informed Trees (BIT*): Informed asymptotically optimal anytime search," *International Journal of Robotics Research*, vol. 39, no. 5, pp. 543–567, 2020.

[5] M. P. Strub and J. D. Gammell, "Adaptively Informed Trees (AIT*) and Effort Informed Trees (EIT*): Asymmetric bidirectional sampling-based path planning," *International Journal of Robotics Research*, vol. 41, no. 4, pp. 390–417, 2022.

[6] L. Zhang, S. Wang, K. Cai, Z. Bing, F. Wu, C. Wang, S. Haddadin, and A. Knoll, "APT*: Asymptotically optimal motion planning via adaptively prolated elliptical r-nearest neighbors," *IEEE Robotics and Automation Letters*, 2025.

[7] L. Zhang *et al.*, "Flexible Informed Trees (FIT*): Adaptive batch-size approach in informed sampling-based path planning," in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 3146–3152.

[8] N. Ratliff, M. Zucker, J. A. Bagnell, and S. Srinivasa, "CHOMP: Gradient optimization techniques for efficient motion planning," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2009, pp. 489–494.

[9] M. Zucker *et al.*, "CHOMP: Covariant Hamiltonian optimization for motion planning," *International Journal of Robotics Research*, vol. 32, no. 9–10, pp. 1164–1193, 2013.

[10] B. Kim, T. T. Um, C. Suh, and F. C. Park, "Tangent bundle RRT: A randomized algorithm for constrained motion planning," *Robotica*, vol. 34, no. 1, pp. 202–225, 2016.

[11] P. H. Kyaw and J. Kelly, "Geometry-aware sampling-based motion planning on Riemannian manifolds," *arXiv preprint arXiv:2602.00992*, 2026.

[12] Y. Zhang, J. Wu, W. Zhang, and Y. Wang, "Bi-AM-RRT*: A bidirectional attractor-guided sampling-based motion planning via assisting metric," *IEEE Transactions on Intelligent Vehicles*, 2023.

[13] H. Klein, N. Jaquier, A. Meixner, and T. Asfour, "On the design of region-avoiding metrics for collision-safe motion generation on Riemannian manifolds," in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.

[14] C.-A. Cheng, M. Mukadam, J. Issac, S. Birchfield, D. Fox, B. Boots, and N. Ratliff, "RMPflow: A computational graph for automatic motion policy generation," in *Proc. Workshop on the Algorithmic Foundations of Robotics (WAFR)*, 2018.

[15] H. Beik-Mohammadi, S. Hauberg, G. Arvanitidis, G. Neumann, and L. Rozo, "Reactive motion generation on learned Riemannian manifolds," *arXiv preprint arXiv:2203.07761*, 2023.

[16] E. Coumans and Y. Bai, "PyBullet, a Python module for physics simulation for games, robotics and machine learning," http://pybullet.org, 2016–2021.